

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



**ST. ANDREW'S JUNIOR COLLEGE
2017 JC2 PRELIMS**

H2 BIOLOGY

9744/4

Paper 4: Practical Exam

Friday

25th August 2017

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Shift
Lab

INFORMATION TO CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 21
2	/19
3	/15
Total	/55

This document consists of **18** printed pages.

[Turn over

IMPORTANT:

Candidates with access to microscope at the start of the paper are given the **first 1h 15 min** to use it. Please answer **QUESTION 2(b)** within this time frame.

Candidates with no access to microscope at the start of the paper will be given access **1h 15min after the start of the paper**. You may proceed with **QUESTION 1** first.

QUESTION 1**Investigation into the effect of enzyme concentration on the hydrolysis of starch**

The enzyme amylase catalyses the hydrolysis of starch.

You are required to investigate the effect of the concentration of amylase on the time taken to completely hydrolyse starch.

Iodine solution turns from yellowish brown to blue-black when starch is present. The time taken for the complete hydrolysis of starch can be found by removing a sample of an amylase and starch mixture at regular time intervals, and adding it to a drop of iodine solution. The starch has been completely hydrolysed when the iodine solution remains yellowish brown after adding the sample. This is the end-point. In judging the end-point, specks of blue-black in an otherwise yellowish brown solution can be ignored.

You are provided with:

- 40.0 cm³ of 2.0% amylase solution, labelled **E**, which is an irritant,
- 100.0 cm³ of distilled water, labelled **W**,
- 40.0 cm³ of starch solution, labelled **S**,
- Iodine solution, labelled **iodine**, which is a stain.

Read steps 1 to 9 before starting.

Proceed as follows:

You are required to prepare different concentrations of the amylase solution and set up a control.

1 Carry out **simple** dilutions of the amylase solution, **E**, to obtain a range of concentrations in which the concentration of the amylase is reduced by 0.4% between each successive dilution.

Prepare 10.0 cm³ for each concentration of amylase solution, **using the plastic containers provided**.

Complete **Table 1.1** to show how you will prepare the different concentrations of amylase solution. [2]

Table 1.1

Concentration of amylase solution / %	volume of E / cm ³	volume of W / cm ³
2.0		

2 Prepare a suitable control for this investigation.

Describe the control that you have prepared. Explain the rationale of the control.

.....

[2]

You are required to investigate the effect of different amylase concentrations on the time taken to completely hydrolyse starch. You will test samples from mixtures of a starch solution and different concentrations of amylase solution at a chosen time interval until each end-point is reached, up to a maximum of **180 seconds**.

3 Decide on a suitable time interval at which samples from each mixture of amylase solution and starch solution will be tested for complete hydrolysis of starch.

State the chosen time interval with a reason for this choice.

Time interval

Reason

[1]

4 The enzyme's optimum temperature is 35°C. Using the hot and tap water provided, set up a water-bath in the beaker labelled **water-bath**, so that you can maintain this temperature throughout the investigation.

5 Put uniform-sized drops of iodine solution on the white tile, labelled with the times that a sample from each mixture of amylase solution and starch solution will be removed and tested, as shown in **Fig. 1.1**.

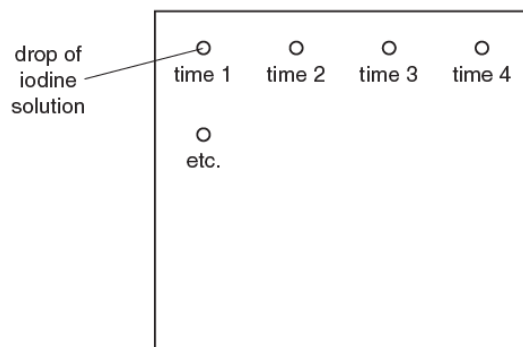


Fig. 1.1

6 Put 3.0 cm³ of **S** into a test-tube and 2.0 cm³ of 2.0% amylase solution into a separate test-tube. Put the test-tubes into the water-bath for at least one minute in order to equilibrate to 35°C.

7 The reaction will start as soon as **S** and the amylase solution are mixed.

Add the starch solution, **S**, to the 2.0% amylase solution, and start timing immediately. Using a Pasteur pipette, remove a sample of the mixture at the first chosen time and add **one drop** to the first drop of iodine solution on the white tile. Continue removing and testing samples at the chosen time interval until the end-point is reached, up to a maximum of **180 seconds**. *Make sure that the mixture or enzyme and starch are maintained at 35°C throughout the experiment.*

8 Repeat steps **5 – 7** to collect the result for each of the other concentrations of amylase solution and the control that you have prepared. Record '**more than 180**' for any mixtures that have not reached the end-point by 180 seconds.

9 Use the space below to record your results.

[4]

10 Explain how the concentration of amylase affects the rate of hydrolysis of starch.

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

11 Temperature was one variable which was controlled in this investigation.

Identify one variable that affects enzyme reactions, which was not controlled in this investigation.

.....[1]

12 Suggest how you would control this variable.

.....[1]

13 For a biotechnological process involving an enzyme to work most efficiently, the enzyme must work at its maximum rate, R.

An enzyme can be used to catalyse the conversion of ethanol (substrate) to acetaldehyde (product).

The effect of the concentration of ethanol (A) on the maximum rate of the production of acetaldehyde (R), is shown in **Table 1.2**.

Table 1.2

Concentration of ethanol (A) / mol dm ⁻³	Maximum rate (R) / min ⁻¹
0.00800	0.0700
0.0150	0.110
0.0500	0.170
0.100	0.220
0.300	0.270

A linear graph can be drawn by plotting 1/R against 1/A.

This can be used to find R (maximum rate of production) for any particular ethanol concentration in this range.

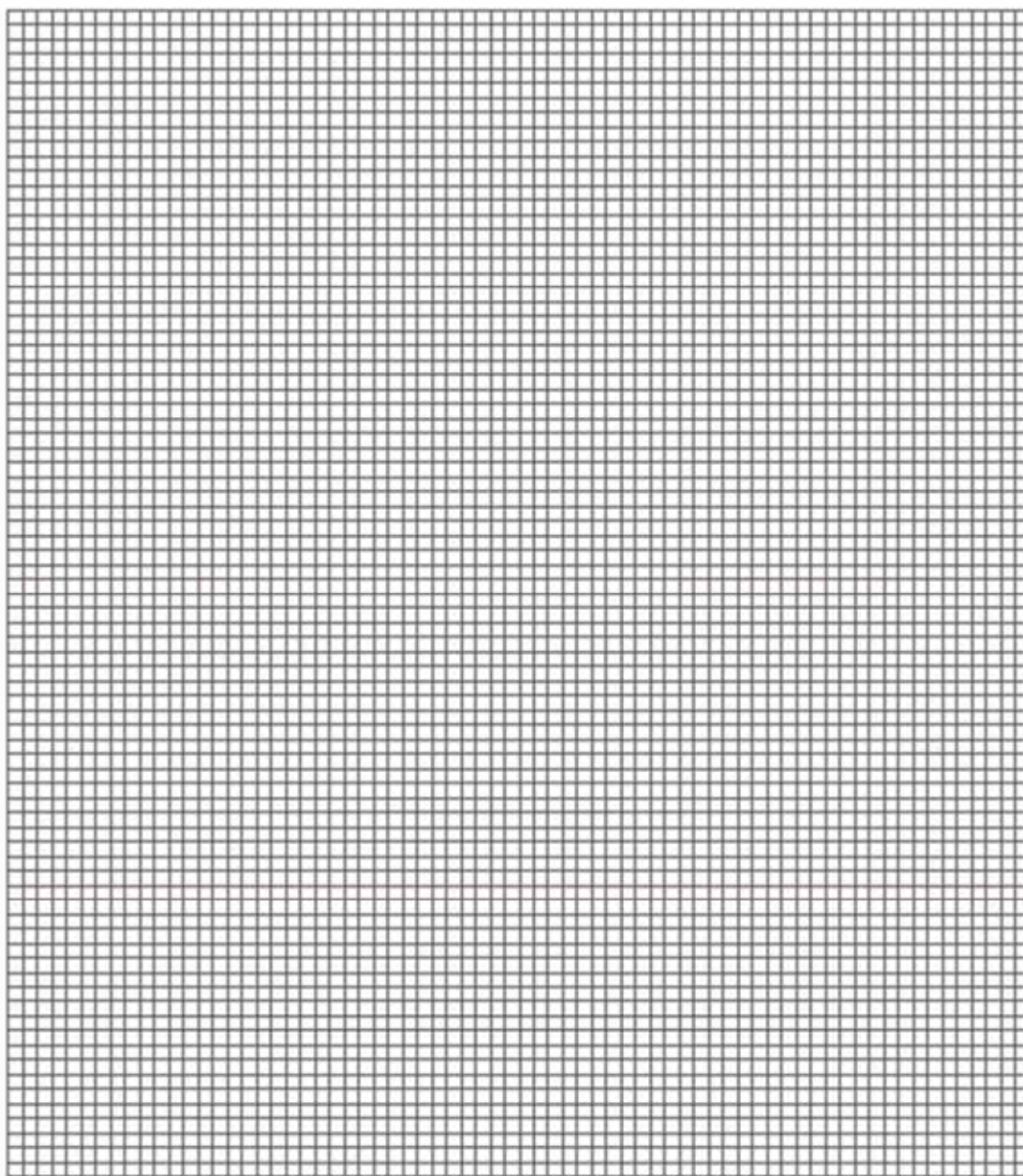
Complete **Table 1.3** for the values of A and R in the last row of Table 1.2, by calculating 1/A and 1/R to the appropriate number of decimal places. [1]

Table 1.3

1/A / mol ⁻¹ dm ³	1/R / min
125.0	14.3
66.7	9.1
20.0	5.9
10.0	4.6

14 Using the data from **Table 1.3**, draw a graph on the grid provided.

[4]



15 Find the maximum rate of production (R) which would be achieved if the ethanol concentration (A) was 0.1 mol dm^{-3} .

Show clearly how you obtained R.

R = min^{-1} [2]

[TOTAL : 21]

QUESTION 2

Fig 2.1 is a photomicrograph of a stained transverse section through part of a leaf from a different type of plant.

You are not expected to be familiar with this specimen.

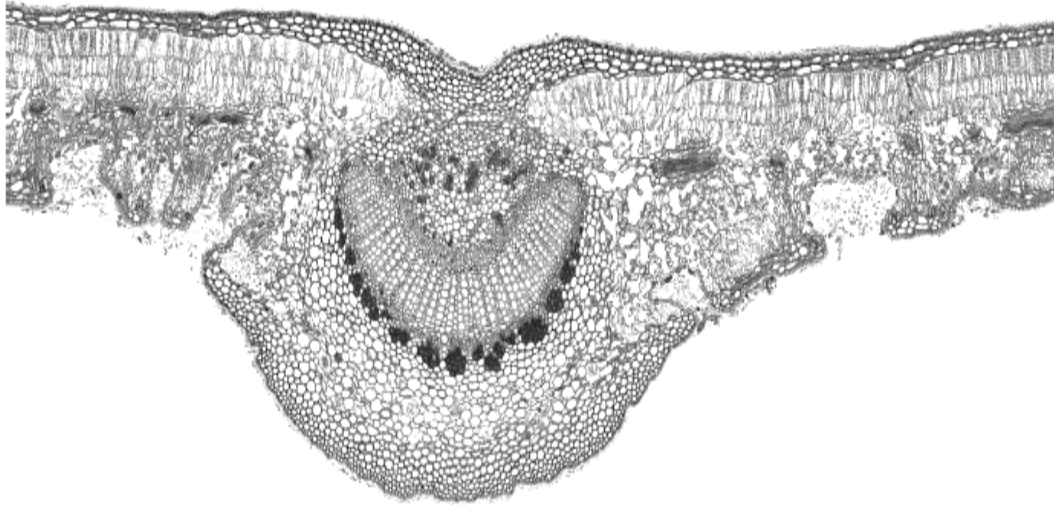


Fig. 2.1

(a) Draw a large **plan diagram** of Fig. 2.1 in the space provided below. Please refer to the coloured photo micrograph provided on the student's bench. [2]

REMINDER:

Candidates with access to microscope at the start of the paper are given the **first 1h 15 min** to use it. Please answer **QUESTION 2(b)** within this time frame.

(b) You are required to measure the diameter of the field of view using the clear plastic ruler.

Proceed as follows:

1. Put the clear plastic ruler on the stage of the microscope and view the scale lines using low power ($\times 100$).
2. Measure the diameter of the field of view and record this in **(b)(i)**.

(i) Diameter of the field of viewmm [1]

Fig. 2.2 is the same photomicrograph as in **Fig. 2.1** showing the field of view at the same magnification as the field of view you have just measured.

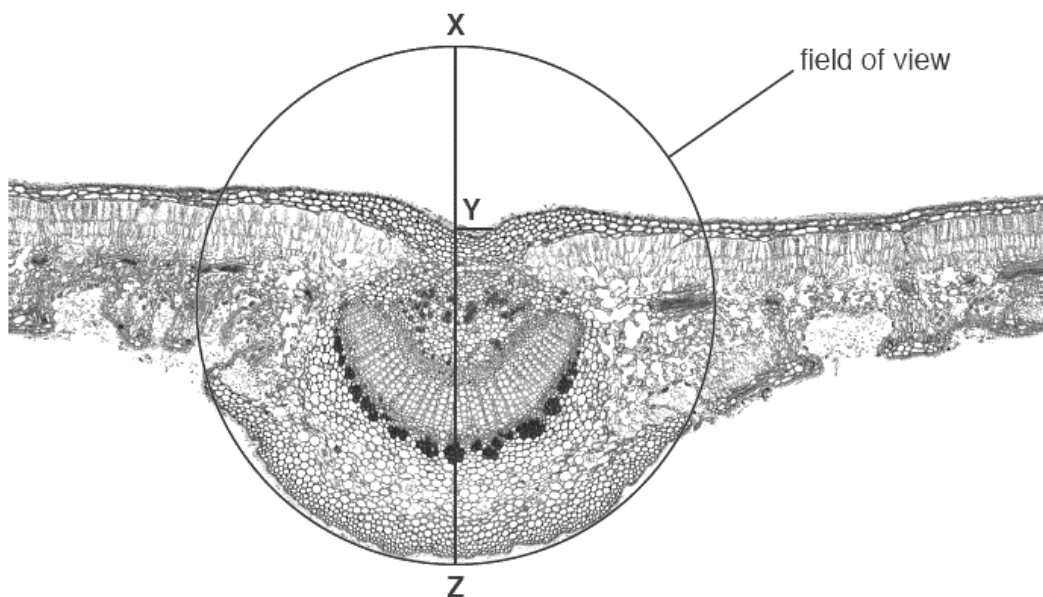


Fig. 2.2

(ii) Using appropriate measurements, calculate the fraction of the diameter of the field of view occupied by the leaf in **Fig. 2.2** along the line X–Z.

fraction of diameter of field of view [1]

(iii) Using your answers to **(b)(i)** and **(b)(ii)** calculate the depth of the midrib, as shown by line **Y–Z**. Give your answer to the nearest μm .
You may lose marks if you do not show your working.

..... μm [2]

(iv) A student used a clear plastic ruler to measure the field of view of a microscope. The student replaced the ruler with a slide of a leaf and **estimated** the diameter of the midrib. Using these results the student calculated the actual diameter of the midrib.

State how this student could have modified their method to obtain a more accurate result. State the apparatus the student would use and describe the method.

apparatus

.....

method

.....

.....

.....[3]

(c) One technique used for studying antigen-antibody reactions is immunodiffusion.

Wells are cut into an agar support medium to contain antigens and antibodies. Antibodies and antigens diffuse out of the wells into the agar. If an antigen meets a complementary antibody a reaction occurs causing a band of precipitate to appear.

Fig. 2.3 shows the results of an immunodiffusion test with known antigens **P** and **Q** and the antibodies to these antigens.

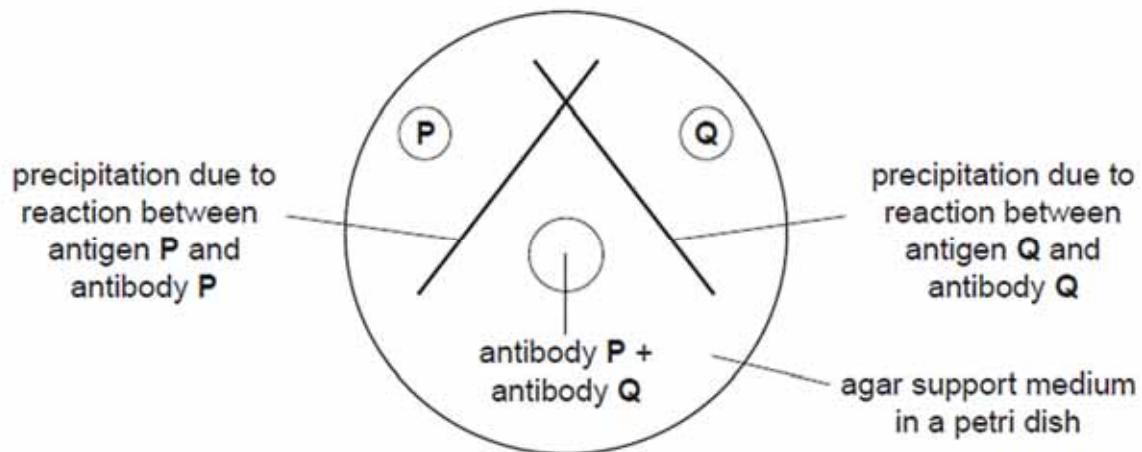


Fig. 2.3

In an investigation, the serum from two test organisms was tested for the presence of antibodies to specific antigens. Both organisms had been previously exposed to both antigens. The serum was placed in wells at the edge of the petri dish and the antigens in a central well.

Fig. 2.4 shows the test set-up.

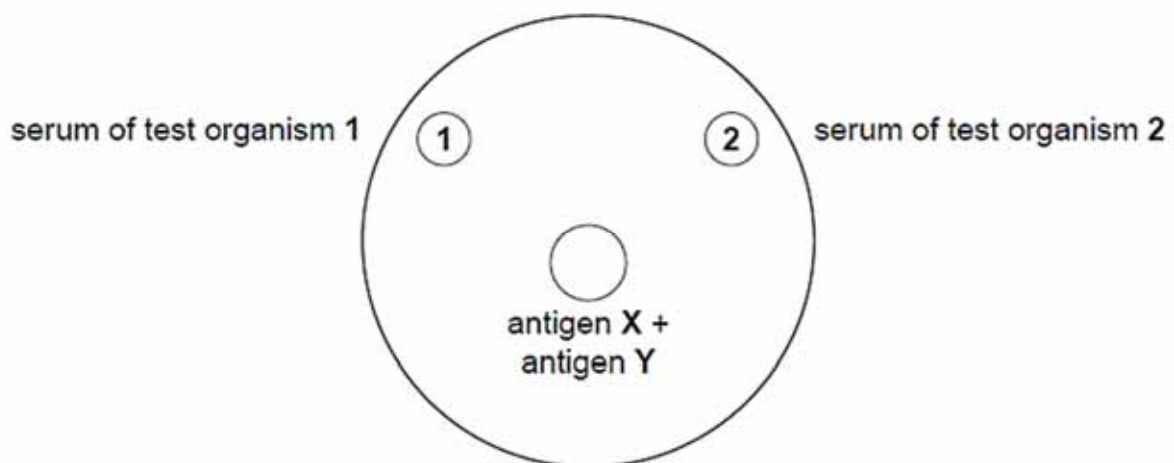


Fig. 2.4

(i) Suggest **one** variable that must be controlled in this procedure.

.....[1]

(ii) State the independent variable in this investigation.

.....[1]

(iii) Both test organisms had antibodies against antigen **X**, but only organism **2** had antibodies against antigen **Y**.

On Fig. 2.4 draw lines to represent where precipitation might have occurred for both organisms. [2]

(iv) Suggest one disadvantage of immunodiffusion for detecting antigens.

.....
.....[1]

A **naturally occurring** mutant of *Plasmodium* sp. has been tested for use as a 'whole organism' vaccination against malaria. The mutant organism develops normally in mosquito vectors and infects the salivary glands in the same way as non-mutant wild type *Plasmodium* sp. In mice, the mutant infects liver cells but does not multiply and cannot enter red blood cells.

Trials using mice were carried out and the effectiveness of the mutant organism as a vaccine tested by injecting non-mutant wild type *Plasmodium* sp. into vaccinated and non-vaccinated mice.

Table 2.1 shows the results of investigations in mice using the mutant *Plasmodium* sp.

Table 2.1

test group	number of mutant <i>Plasmodium</i> cells given to the mice			percentage of mice not infected by wild type <i>Plasmodium</i> sp.
	first inoculation	first booster inoculation	second booster inoculation	
1	0	0	0	0
2	50 000	25 000	25 000	100
3	10 000	10 000	10 000	100
4	10 000	10 000	0	70

(v) Suggest the purpose of including each of the following test groups.

group 1
.....
groups 2 and 3
.....
group 4
.....[3]

(vi) Using the information in the question, outline a procedure that might be used to obtain mutant *Plasmodium* sp. to use in the vaccination trials.

.....
.....
.....
.....[2]

[TOTAL : 19]

QUESTION 3: PLANNING QUESTION

Effect of citrate on rate of respiration

Enzymes catalysing essentially irreversible reactions are potential sites of control in cellular respiration. One of these enzymes is phosphofructokinase, which can be regulated by the reversible binding of citrate to its allosteric site. (Citrate is produced as an intermediate compound during Krebs cycle.)

Using this information and your own knowledge, design an experiment to determine the effect of citrate concentration on the rate of cellular respiration.

You must use:

- 10 mM citrate,
- purified homogenate of enzymes found in the cytosol,
- 5% glucose solution,
- pH buffer,
- distilled water,
- Benedict's solution,
- apparatus shown in **Fig. 3.1**, can be used to separate proteins from ions and disaccharides,

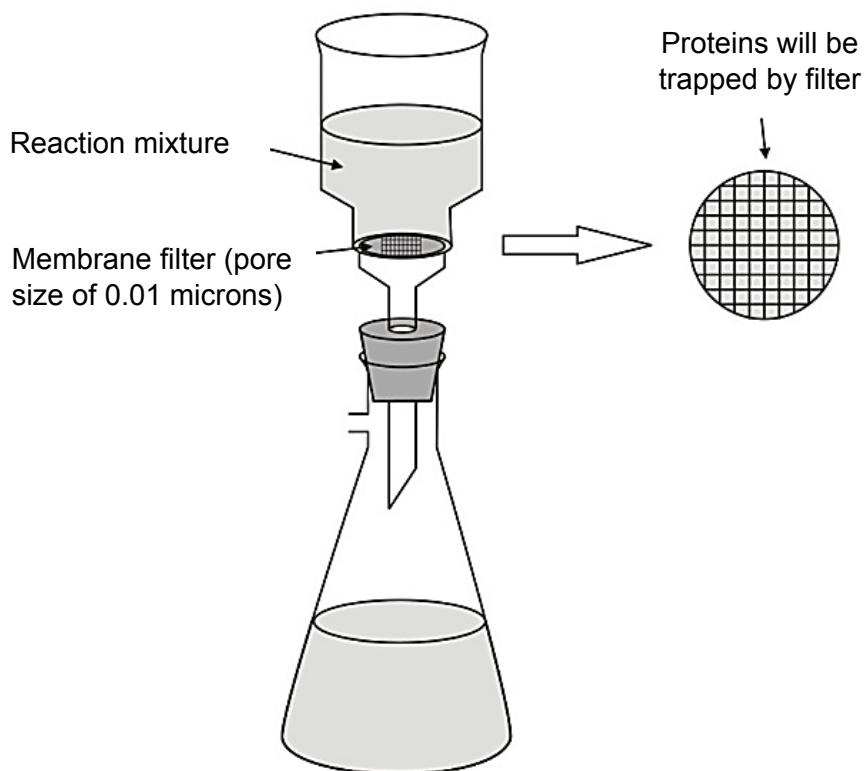


Fig. 3.1

- syringes,
- white card,

- stopwatch,
- thermometer,
- bunsen burner with tripod, gauze and bench mat,
- thermostatically controlled water bath,
- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 15]

